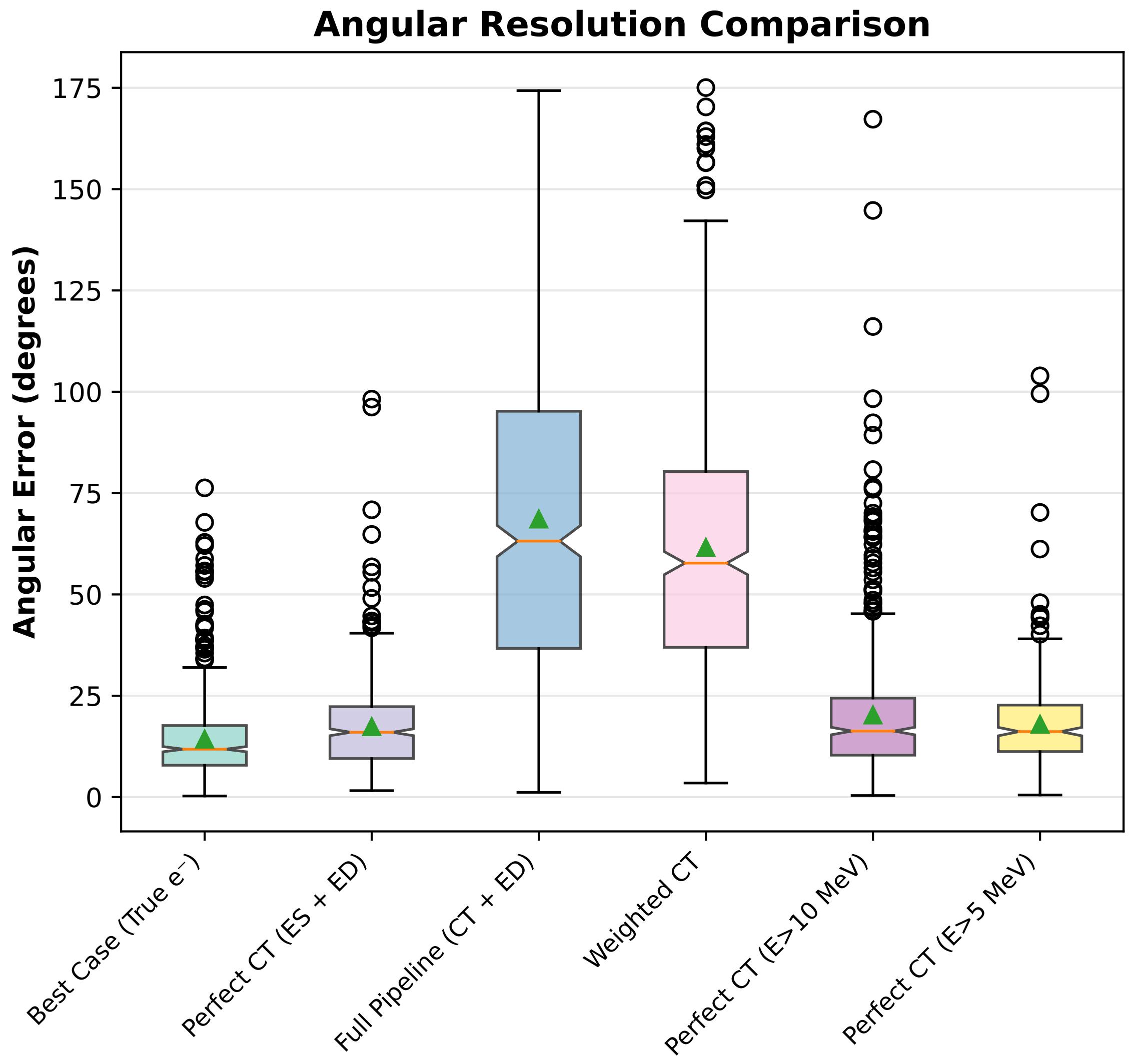
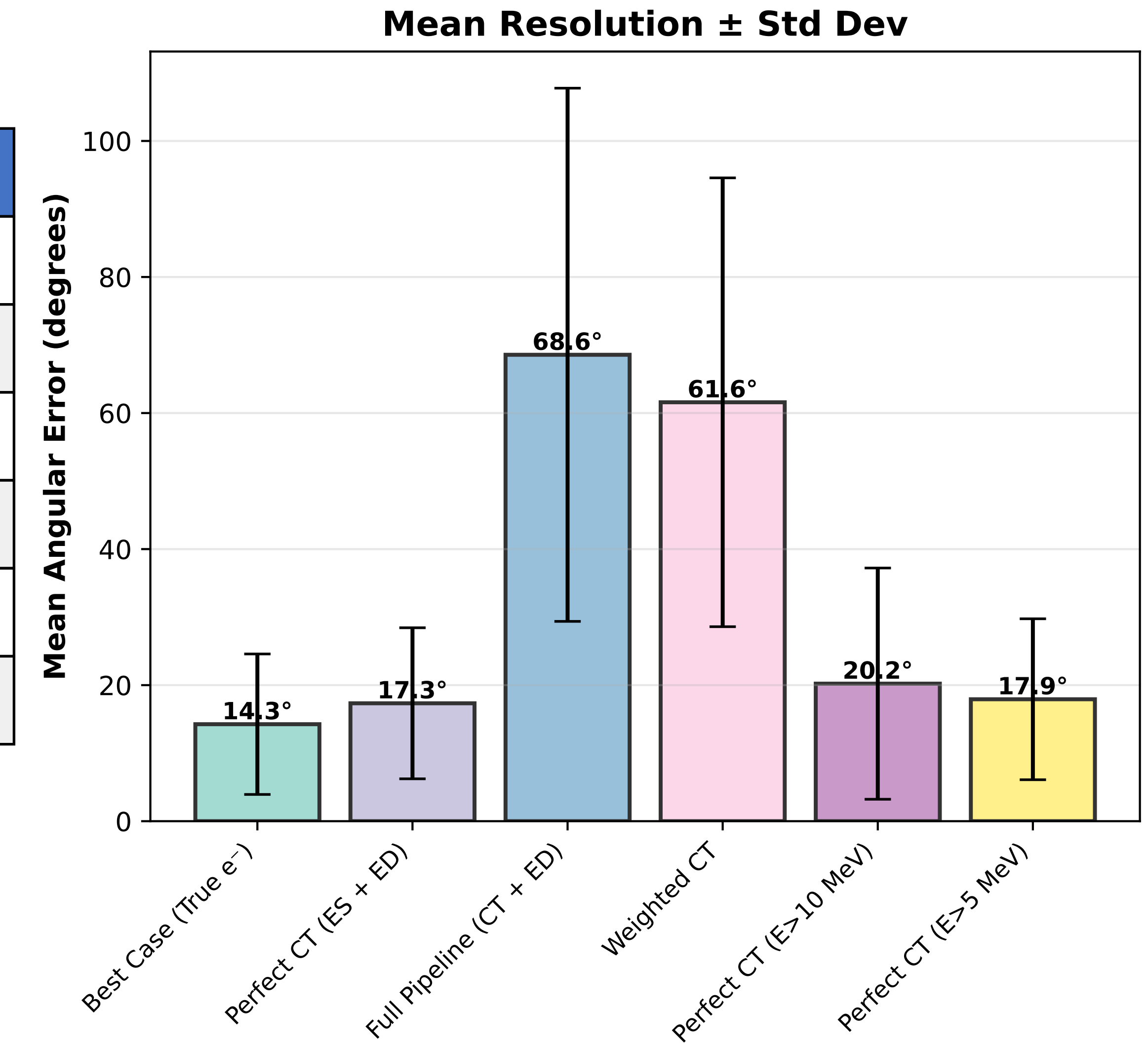
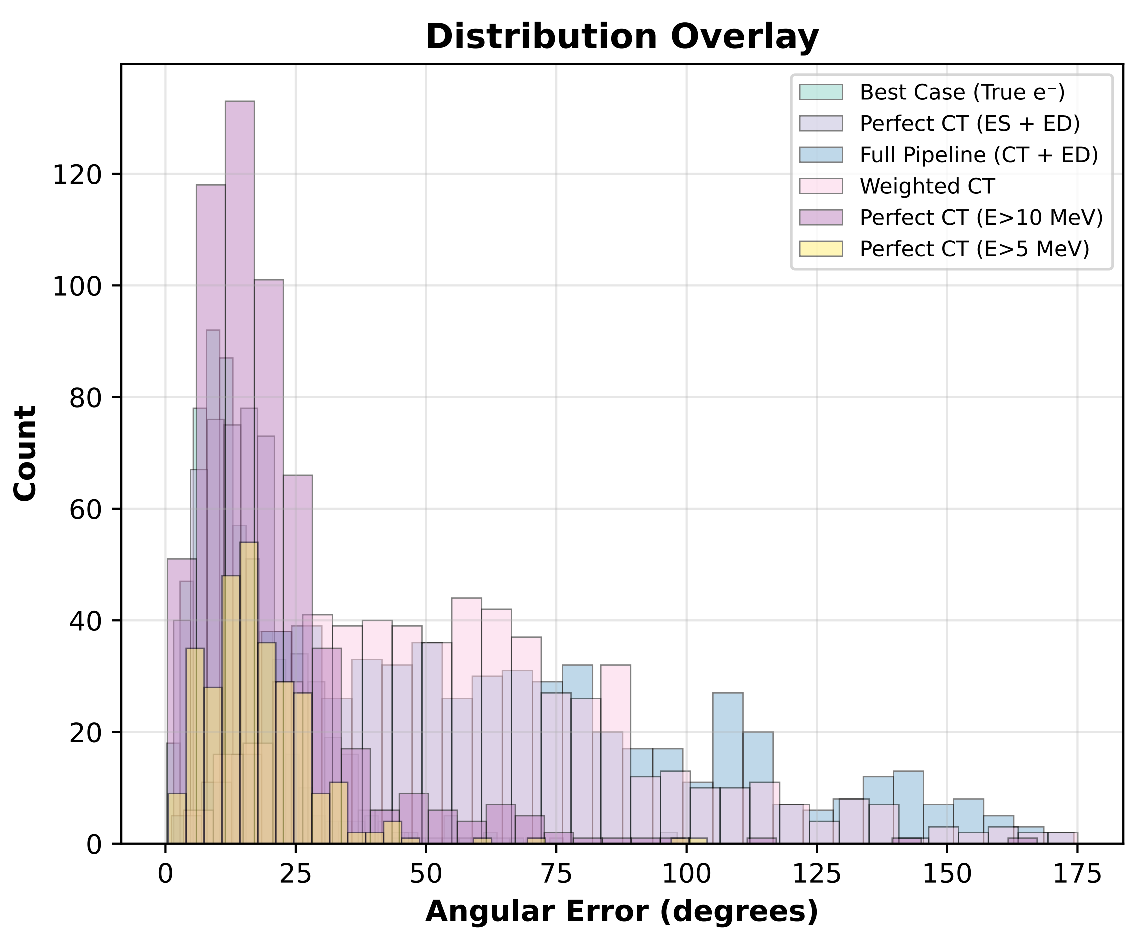
Emcee 6-Scenario Aggregate Analysis

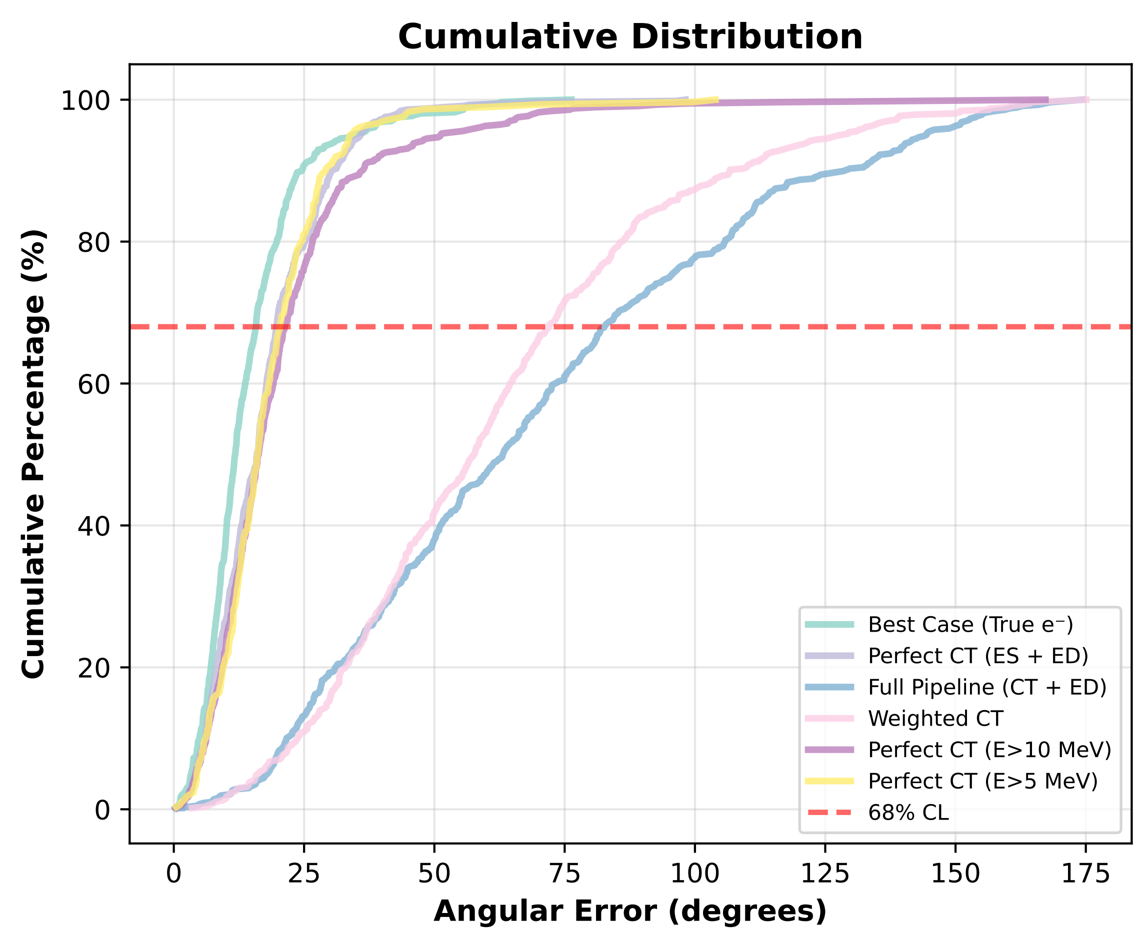


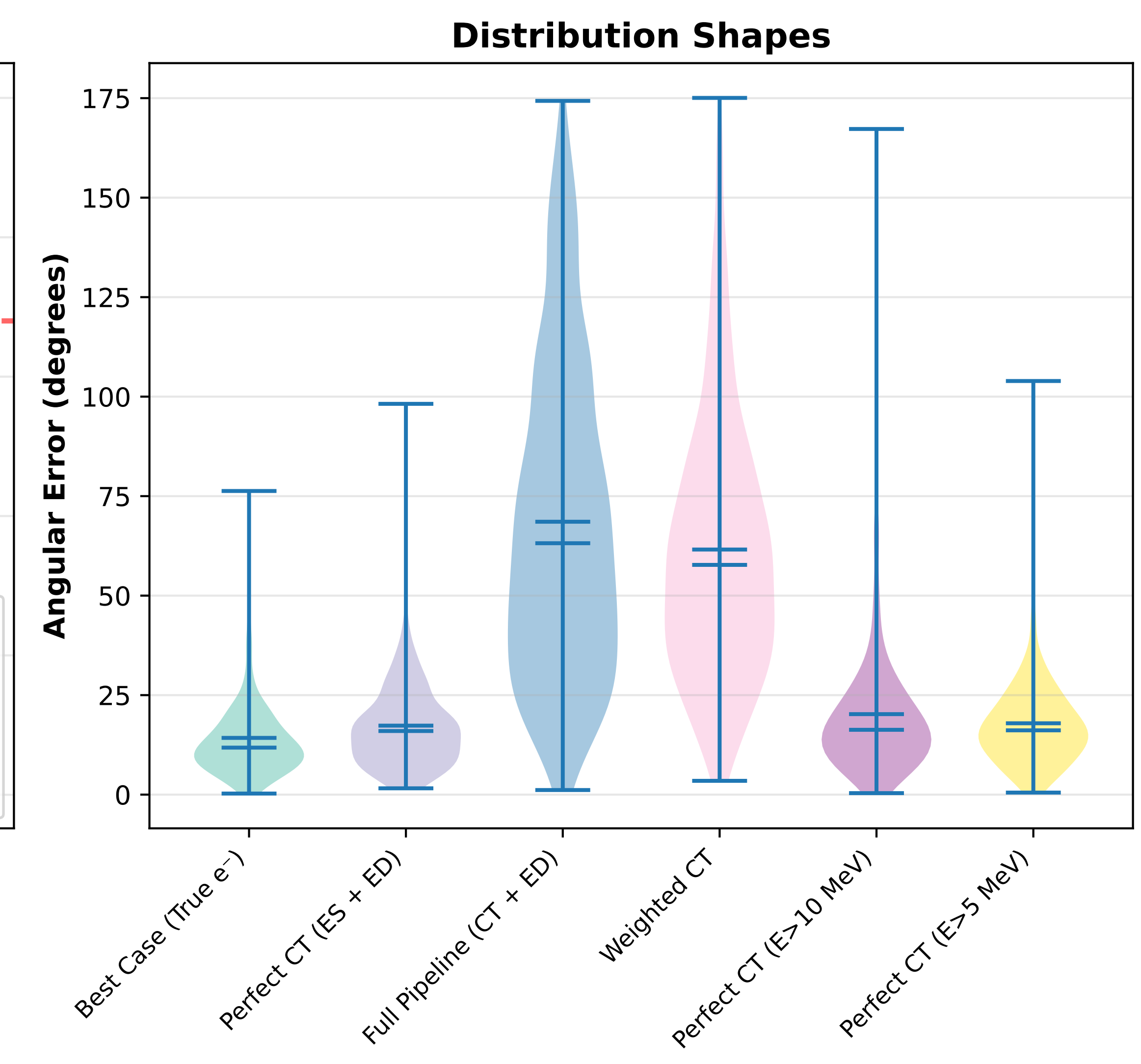
Summary Statistics

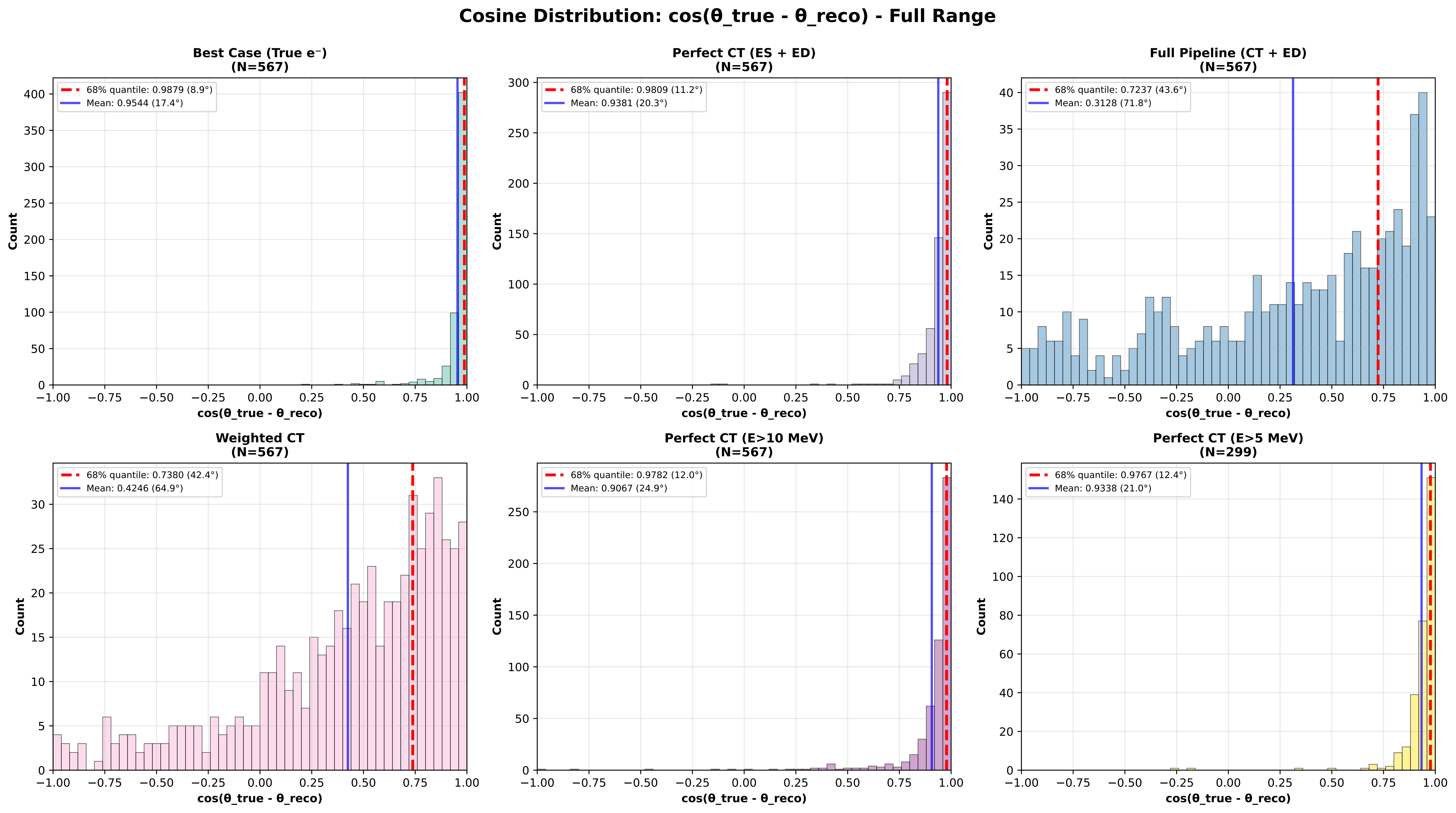
Scenario	N	68% Q	Median	Mean
Best Case (True	567	15.78°	11.81°	14.26°
Perfect CT (ES	567	19.69°	15.98°	17.33°
Full Pipeline (C	567	82.66°	63.17°	68.57°
Weighted CT	567	71.97°	57.73°	61.58°
Perfect CT (E>	567	21.34°	16.29°	20.22°
Perfect CT (E>	299	20.36°	16.15°	17.93°

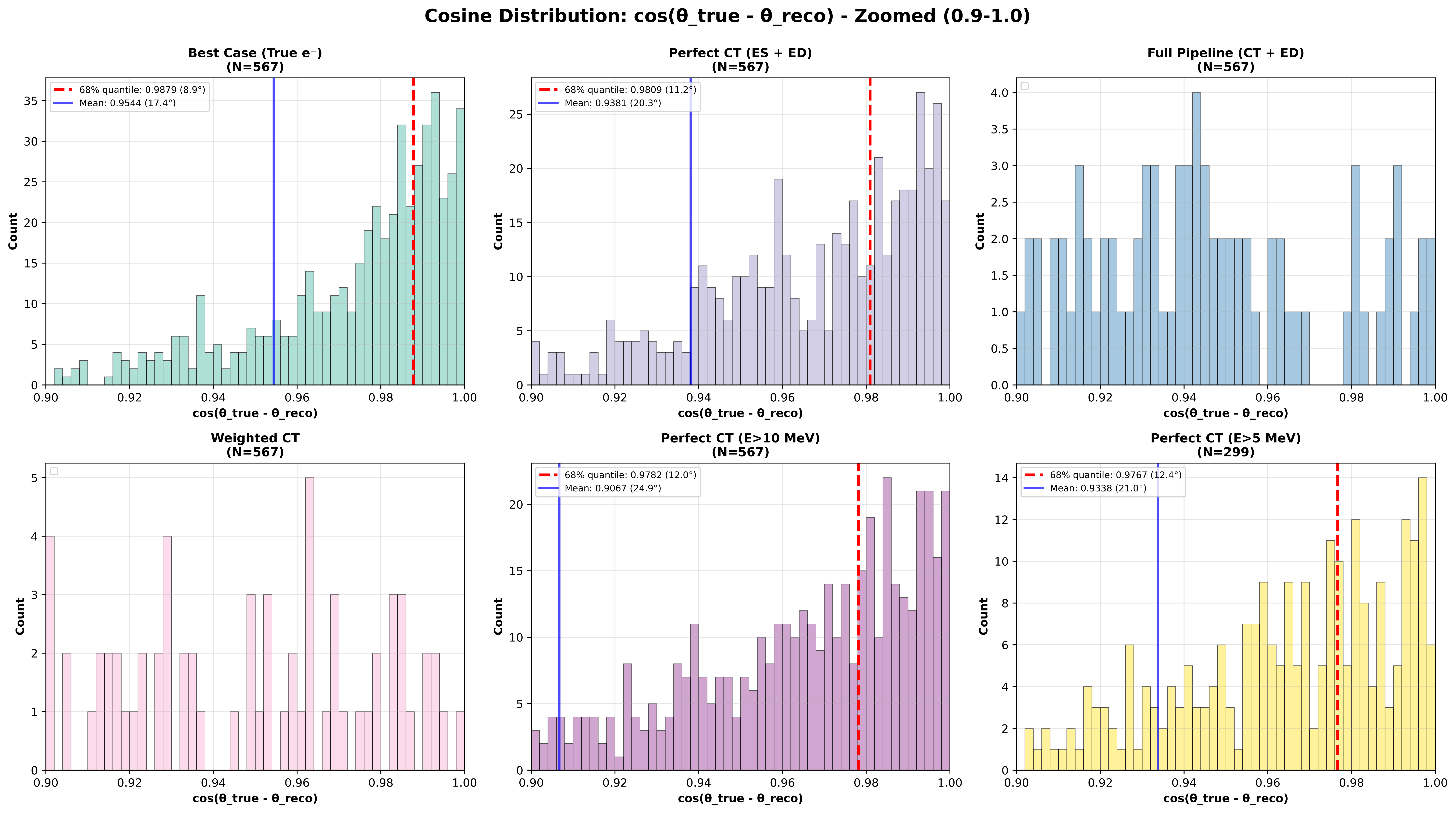









Cosine Distribution: cos(θ_true - θ_reco) - Full Range
Best Case (True e⁻)
(N=567)
68% quantile: 0.9879 (8.9°)
Mean: 0.9544 (17.4°)
Perfect CT (ES + ED)
(N=567)
68% quantile: 0.9809 (11.2°)
Mean: 0.9381 (20.3°)
Full Pipeline (CT + ED)
(N=567)
68% quantile: 0.7237 (43.6°)
Mean: 0.3128 (71.8°)
Weighted CT
(N=567)
68% quantile: 0.7380 (42.4°)
Mean: 0.4246 (64.9°)
Perfect CT (E>10 MeV)
(N=567)
68% quantile: 0.9782 (12.0°)
Mean: 0.9067 (24.9°)
Perfect CT (E>5 MeV)
(N=299)
68% quantile: 0.9767 (12.4°)
Mean: 0.9338 (21.0°)
Count
cos(θ_true - θ_reco)



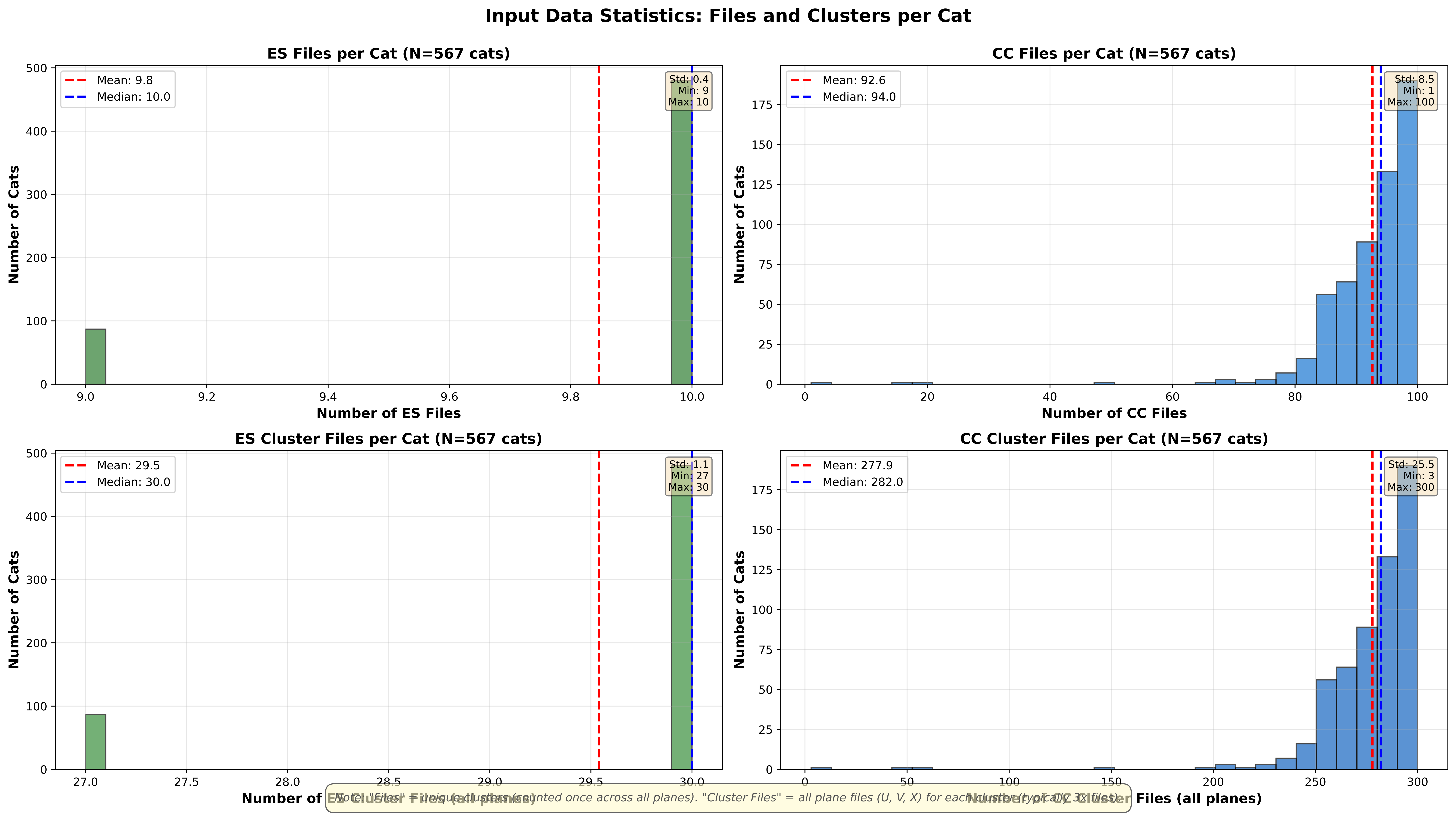
Cosine Distribution: cos(θ_true - θ_reco) - Zoomed (0.9-1.0)
Best Case (True e⁻)
(N=567)
68% quantile: 0.9879 (8.9°)
Mean: 0.9544 (17.4°)
Perfect CT (ES + ED)
(N=567)
68% quantile: 0.9809 (11.2°)
Mean: 0.9381 (20.3°)
Full Pipeline (CT + ED)
(N=567)
Weighted CT
(N=567)
Perfect CT (E>10 MeV)
(N=567)
68% quantile: 0.9782 (12.0°)
Mean: 0.9067 (24.9°)
Perfect CT (E>5 MeV)
(N=299)
68% quantile: 0.9767 (12.4°)
Mean: 0.9338 (21.0°)
Count
cos(θ_true - θ_reco)

Cluster Usage Across Scenarios

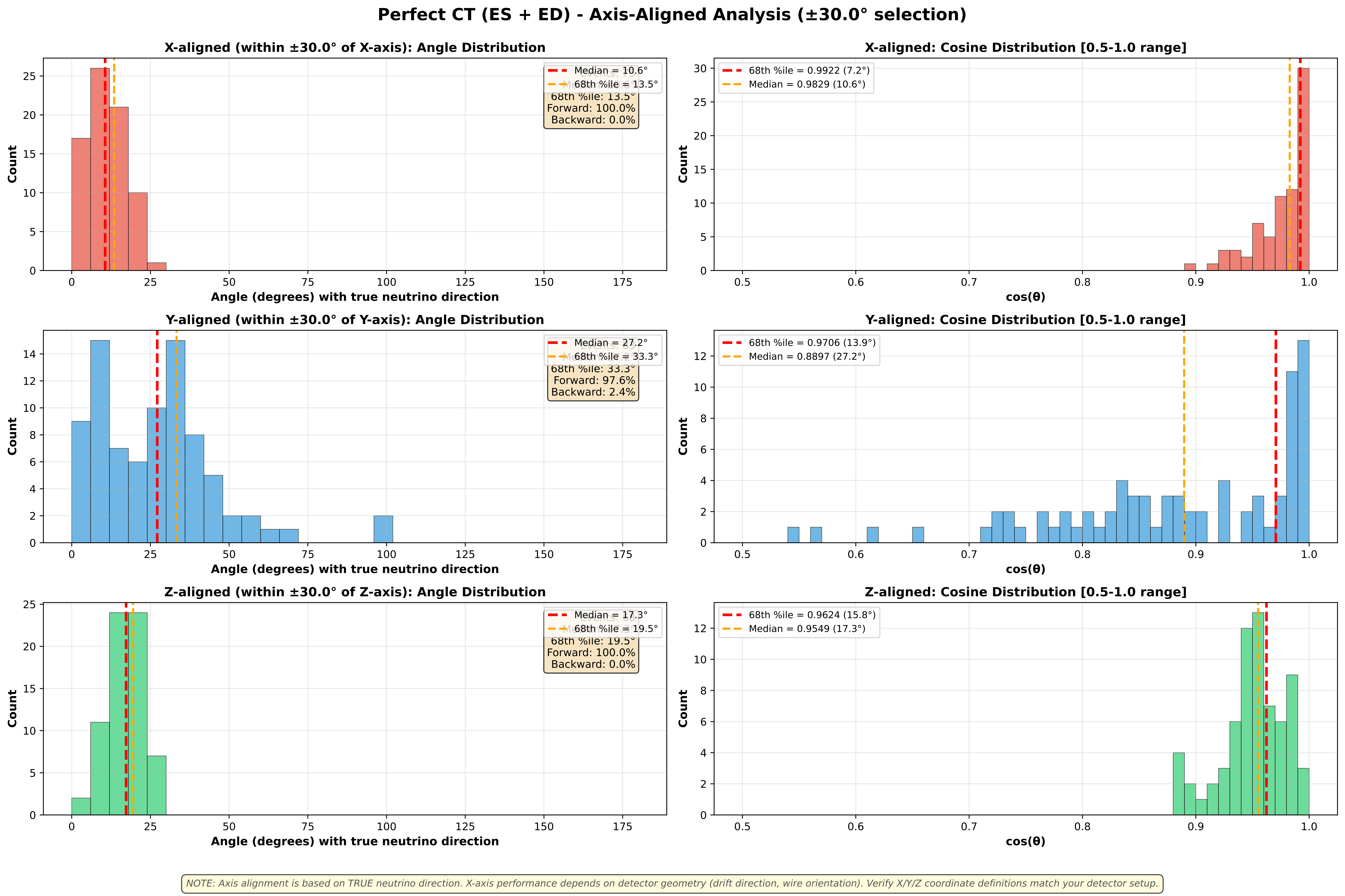
Scenario	N Cats	Input Total	Input ES Main	Value "Clusters Used"	Notes
Best Case (True e⁻)	567	102	10	271	MCMC samples (weighted sum)
Perfect CT (ES + ED)	567	102	10	271	MCMC samples (weighted sum)
Full Pipeline (CT + ED)	567	102	10	3242	MCMC samples (weighted sum)
Weighted CT	567	102	10	3243	MCMC samples (weighted sum)
Perfect CT (E>10 MeV)	567	102	10	147	143.6% retention
Perfect CT (E>5 MeV)	299	101	10	242	MCMC samples (weighted sum)

Note: "Clusters Used" shows different meanings per scenario:
• Best Case, Perfect CT: Number of ES main clusters actually used (after E>3 MeV cut)
• Full Pipeline, Weighted CT: Weighted sum of cluster contributions in MCMC (includes CT probabilities)
• Energy cut scenarios: Number of clusters passing energy threshold

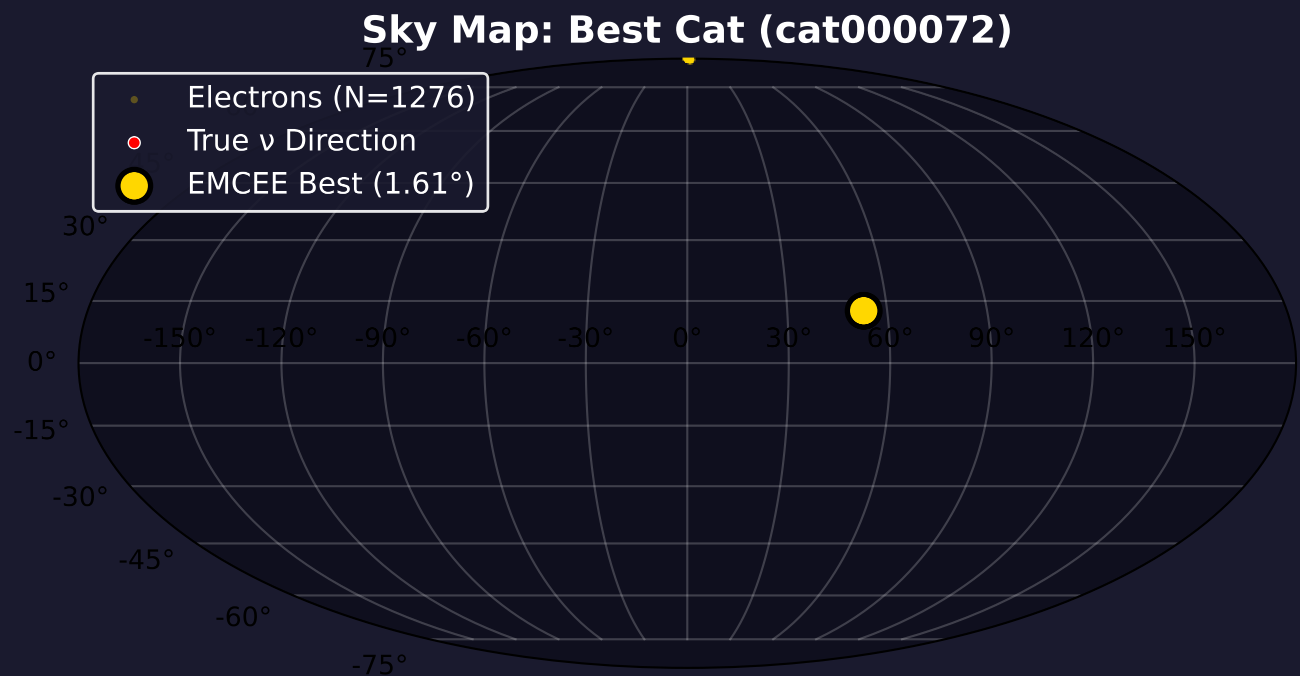
The key insight: Best Case and Perfect CT use ~85-90% of ES main tracks with E>3 MeV.
Full Pipeline shows much higher values due to weighted MCMC sampling, not actual cluster count.

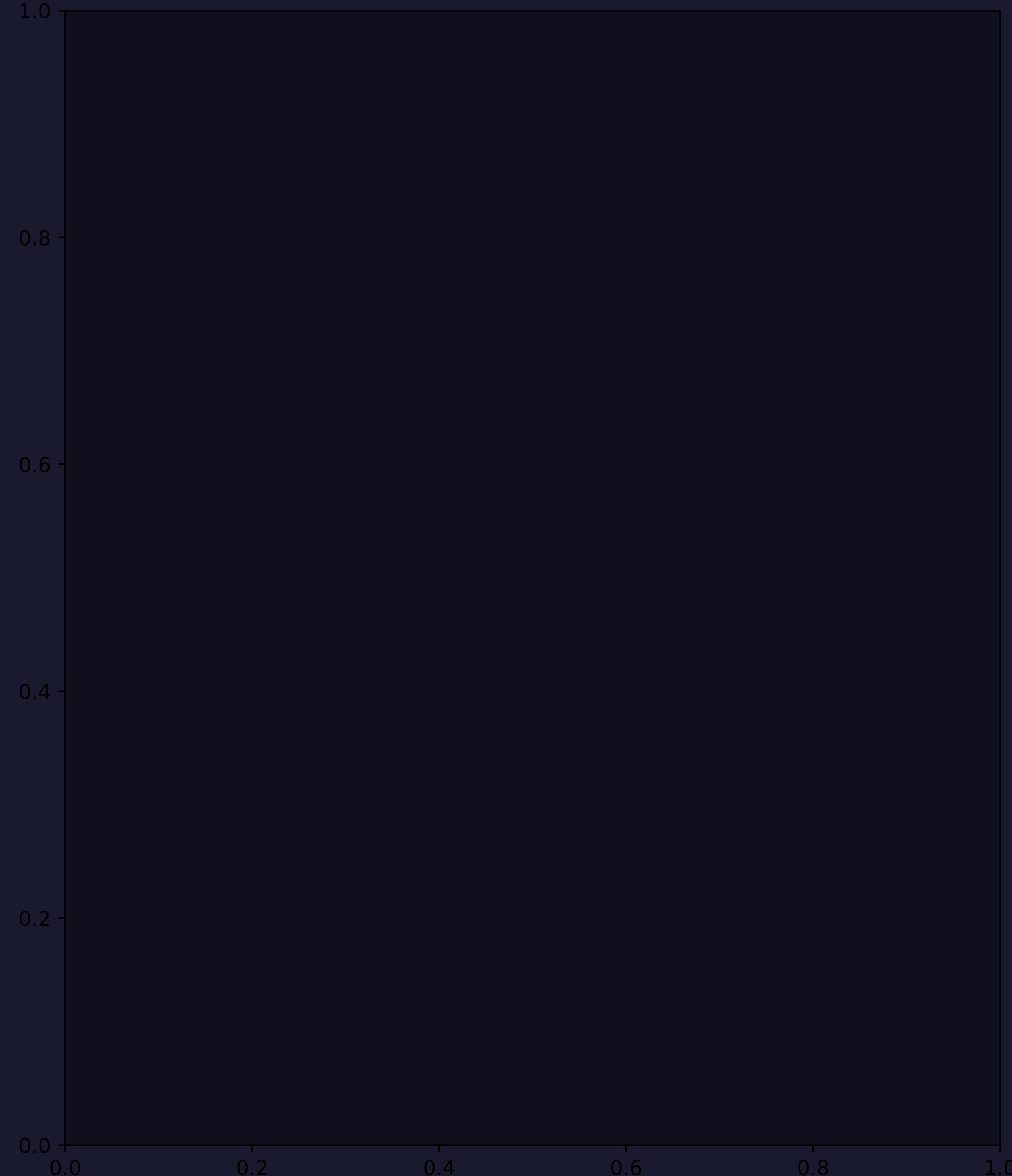


Input Data Statistics: Files and Clusters per Cat
ES Files per Cat (N=567 cats)
Mean: 9.8
Median: 10.0
Std: 0.4
Min: 9
Max: 10
Number of Cats
Number of ES Files
CC Files per Cat (N=567 cats)
Mean: 92.6
Median: 94.0
Std: 8.5
Min: 1
Max: 100
Number of Cats
Number of CC Files
ES Cluster Files per Cat (N=567 cats)
Mean: 29.5
Median: 30.0
Std: 1.1
Min: 27
Max: 30
Number of Cats
Number of ES Cluster Files (all planes)
CC Cluster Files per Cat (N=567 cats)
Mean: 277.9
Median: 282.0
Std: 25.5
Min: 3
Max: 300
Number of Cats
Number of CC Cluster Files (all planes)
Note: "Files" = unique clusters (counted once across all planes). "Cluster Files" = all plane files (U, V, X) for each cluster (typically 3 files).

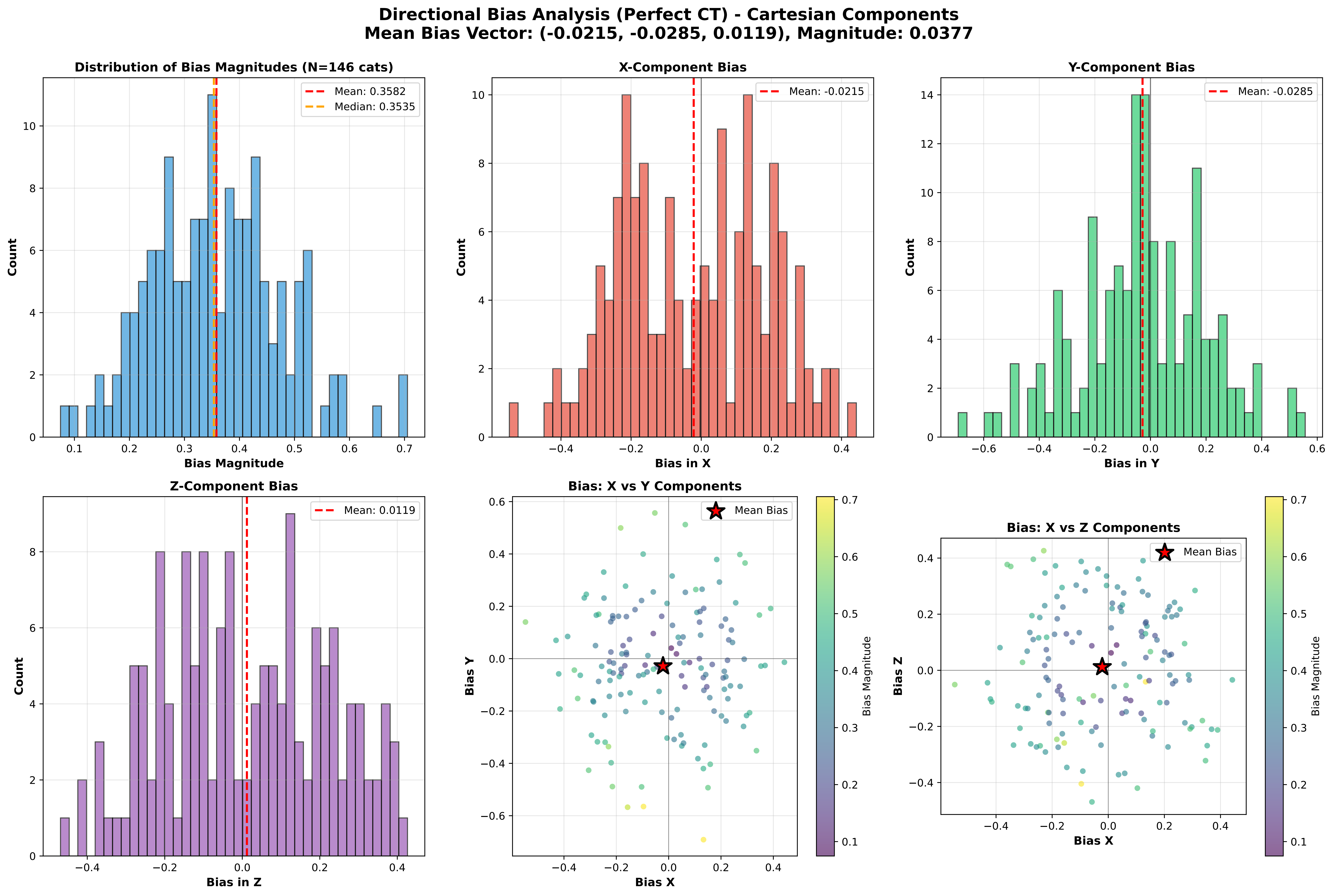

Perfect CT (ES + ED) - Axis-Aligned Analysis (±30.0° selection)
X-aligned (within ±30.0° of X-axis): Angle Distribution
Median = 10.6°
68th %ile = 13.5°
68th %ile: 13.5°
Forward: 100.0%
Backward: 0.0%
Count
Angle (degrees) with true neutrino direction
X-aligned: Cosine Distribution [0.5-1.0 range]
68th %ile = 0.9922 (7.2°)
Median = 0.9829 (10.6°)
Count
cos(θ)
Y-aligned (within ±30.0° of Y-axis): Angle Distribution
Median = 27.2°
68th %ile = 33.3°
68th %ile: 33.3°
Forward: 97.6%
Backward: 2.4%
Count
Angle (degrees) with true neutrino direction
Y-aligned: Cosine Distribution [0.5-1.0 range]
68th %ile = 0.9706 (13.9°)
Median = 0.8897 (27.2°)
Count
cos(θ)
Z-aligned (within ±30.0° of Z-axis): Angle Distribution
Median = 17.3°
68th %ile = 19.5°
68th %ile: 19.5°
Forward: 100.0%
Backward: 0.0%
Count
Angle (degrees) with true neutrino direction
Z-aligned: Cosine Distribution [0.5-1.0 range]
68th %ile = 0.9624 (15.8°)
Median = 0.9549 (17.3°)
Count
cos(θ)
NOTE: Axis alignment is based on TRUE neutrino direction. X-axis performance depends on detector geometry (drift direction, wire orientation). Verify X/Y/Z coordinate definitions match your detector setup.

Neutrino Pointing Reconstruction: cat000072 - Perfect CT Scenario
Yellow: Electrons | Red: True Neutrino Direction

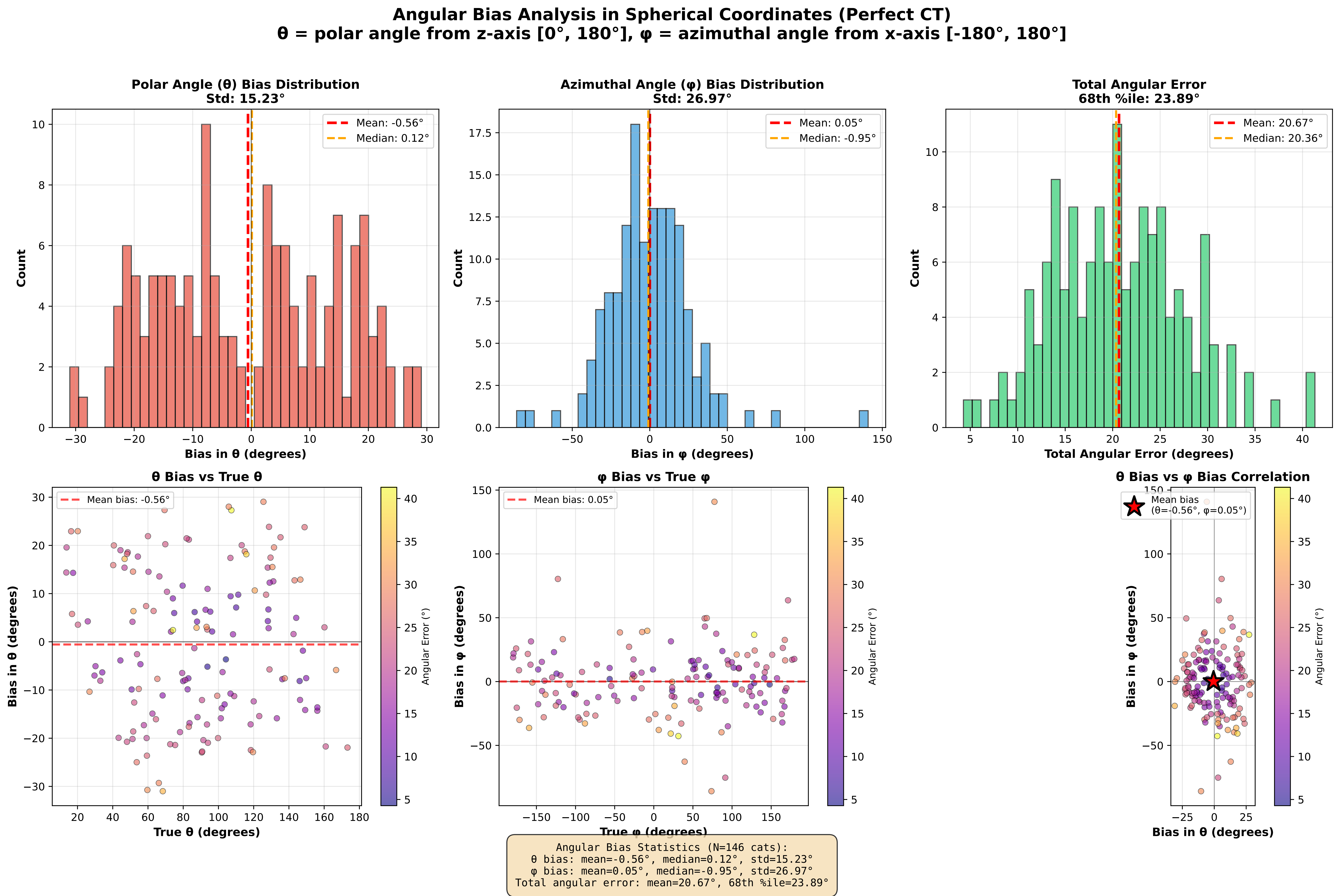




Best cat from EMCEE analysis: cat000072 | Angular Error: 1.61° | 68% Credible Interval (ω_{68}): 0.33° | N electrons: 287



Directional Bias Analysis (Perfect CT) - Cartesian Components
Mean Bias Vector: (-0.0215, -0.0285, 0.0119), Magnitude: 0.0377
Distribution of Bias Magnitudes (N=146 cats)
Mean: 0.3582
Median: 0.3535
Bias Magnitude
Count
X-Component Bias
Mean: -0.0215
Bias in X
Y-Component Bias
Mean: -0.0285
Bias in Y
Z-Component Bias
Mean: 0.0119
Bias in Z
Bias: X vs Y Components
Mean Bias
Bias X
Bias Y
Bias Magnitude
Bias: X vs Z Components
Bias Z


Angular Bias Analysis in Spherical Coordinates (Perfect CT)
θ = polar angle from z-axis [0°, 180°], φ = azimuthal angle from x-axis [-180°, 180°]
Polar Angle (θ) Bias Distribution
Std: 15.23°
Mean: -0.56°
Median: 0.12°
Count
Bias in θ (degrees)
Azimuthal Angle (φ) Bias Distribution
Std: 26.97°
Mean: 0.05°
Median: -0.95°
Bias in φ (degrees)
Total Angular Error
68th %ile: 23.89°
Mean: 20.67°
Median: 20.36°
Total Angular Error (degrees)
θ Bias vs True θ
Mean bias: -0.56°
Bias in θ (degrees)
True θ (degrees)
Angular Error (°)
φ Bias vs True φ
Mean bias: 0.05°
Bias in φ (degrees)
True φ (degrees)
θ Bias vs φ Bias Correlation
Mean bias
(θ=-0.56°, φ=0.05°)
Angular Bias Statistics (N=146 cats):
θ bias: mean=-0.56°, median=0.12°, std=15.23°
φ bias: mean=0.05°, median=-0.95°, std=26.97°
Total angular error: mean=20.67°, 68th %ile=23.89°